Feolu Kolawole

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EDUCATION

Stanford University **Expected Graduation Date: June 2028**
Pursuing a Bachelor's in Computer Science (AI Track and sub-focus in Human-Computer Interaction)
Completed: CS106B: *Data Structures & Algorithms,* CS107: *Computer Org. & Systems*, CS111: *OS's & Concurrency*, MATH51: *Linear Algebra & Multivariable Calculus*, MATH104: *Applied Matrix Theory,* CS231N: *Deep Learning for Computer Vision*

TECHNICAL SKILLS

Programming Languages: Python, Java, C, C#, C++, Swift, Assembly, HTML/CSS, R
Libraries/Frameworks: Tensorflow, Pytorch, SAM, YOLO, SwiftUI, RealityKit, Bootstrap, Dialogflow, React, Numpy, Pandas
Software: Blender, Unity, Adobe Photoshop

PROFESSIONAL EXPERIENCE

Machine Learning and Computer Vision Researcher - HairFlow **Stanford, CA: March 2025 - Present**
Stanford Artificial Intelligence Laboratory (PI - Ron Fedkiw)
- Developed computer vision pipelines for 3D modeling and segmentation, with a focus on enabling human mesh component creation for industry applications in 3D animation, film, and media
- Engineered models for 3D character and avatar development via hair type classification, extraction, and generation

Research Assistant **Palo Alto, CA: April 2025 - Present**
Stanford Human Perception Lab (PI - Khizer Khaderi)
- Enhanced computer vision pipelines by integrating existing SLAM and Point Cloud Segmentation models to transform video data into accurate 3D environmental models at a rate of approximately 72 frames per second
- Facilitated advanced simulations and analyses by seamlessly transferring generated 3D models to NVIDIA Omniverse
- Built a specialized AR application for Snap Spectacles to test users' peripheral vision recognition and reaction time

PROJECTS

Accelerated MRI Reconstruction with SwinUNet Deep Learning **Python | Pytorch | Deep Learning**
Transformer based Deep Learning Architecture for Medical Imaging (View Paper)
- Enabled faster MRI scans for patients by reconstructing high-quality knee MRI images, reducing patient time in the scanner
- Developed an optimized SwinUNet deep learning architecture for accelerated MRI reconstruction, achieving 33.1 dB PSNR and 0.72 SSIM on the fastMRI dataset
- Outperformed baseline U-Net by ~4 dB in PSNR, enhancing MRI image quality and detail preservation from undersampled data

MIT Virtual Reality Hackathon (Healing Heroes) (Winner) **Swift | RealityKit**
Innovative VR-based Healthcare Improvement Simulation
- First place in MIT's VR competition where projects were judged on innovation, impact, and technical complexity
- Designed & Engineered a VR healthcare simulation on the Apple Vision Pro, allowing users to practice CPR & first aid

Supreme Court Case Outcome Prediction Using Bayesian Networks **Python | Numpy | Pandas**
Bayesian Model for Predicting Supreme Court Case Dispositions (View Paper)
- Created a Bayesian decision modeling framework, using observable case attributes to predict Supreme Court case dispositions
- Implemented rejection sampling for inference, achieving a Top-3 accuracy of 73% for prediction amongst 11 possibilities
- 1st place in Stanford CS109 probabilistic modeling competition (~500 students), showcasing expertise in probabilistic methods

Music Recommendation System (View Paper) **Python | SVD | Dimensionality Reduction | PCA**
- Designed a music recommendation system utilizing Singular Value Decomposition (SVD) and Principal Component Analysis (PCA) of Spotify audio features
- Implemented feature projection to identify non-obvious musical similarities, focusing on sound qualities over artist names

SynchroSound **SwiftUI | UIKit | SwiftData | Objective-C**
iOS App for Song Recommendations Based on Facial Mood Recognition
- Built an app that enhances a user's musical experience - deciphers facial expressions and recommends mood-matching songs by integrating SwiftUI, UIKit, and SwiftData with Google Cloud Vision's facial recognition and Spotify's Web API

LEADERSHIP EXPERIENCE

Stanford XR Vice President Of External Affairs **Stanford, CA: September 2024 - Present**
- Co-led successful launch of partnerships with prominent sponsors including Meta, NVIDIA, Amazon, and Snapchat
- Organized immersive technology hackathons with a team, engaging over 300 participants in XR and Spatial Computing projects